

Technology #175: Embed the enhanced physics-informed neural network model into the aircraft lithium■battery management system to continuously predict cell surface temperature during flight and trigger protective actions (e.g., load reduction, active cooling) when predicted temperature approaches runaway thresholds.

Technology Domain: thermal

Mechanism: PROPOSED architecture (no physical system exists): heat source characterization; conduction path / spreader; reject interface; safety limit interlock — realizing: Embed the enhanced physics-informed neural network model into the aircraft lithium■battery management system to continuously predict cell surface temperature during flight and trigger protective actions (e.g., load reduction, active cooling) when predicted temperature approaches runaway thresholds.

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

Next Experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

Buyer Receives: Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

Buyer Must Build: Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

Transfer Posture: Sponsored Validation

Target Buyer: domain-appropriate technology buyer (to be identified by portfolio owner)

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